Macro-systemic Approach to Multi-Asset Investing

Risk Profile Analysis: US Equities (SPY)

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Alpha Capital Jr.
Buy-Side

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Executive summary

This document is part of a macro-driven multi-asset allocation research project. The objective is to design a rules-based framework that translates macroeconomic regimes into systematic portfolio allocations and decisions across a set of assets, targeting risk-adjusted alpha over equity benchmarks.

This section covers individual asset analytics: return distribution, performance, risk, volatility, Sharpe, Sortino, VaR, CVaR, maximum drawdown, and time under water. Each asset is analyzed independently before being integrated into the broader cross-asset and allocation framework.

This report focuses on us equities (SPY). Their role is to drive portfolio returns during favorable growth regimes.

Assets choice

Class	Product	Ticker	First Obs.
Equity	ETF S&P 500	SPY	29-01-1993
Commodities	SPDR Gold Shares	GLD	18-11-2004
Commodities	Commodity Index Tracking Fund	DBC	06-02-2006
Bonds	Treasuries 20Y+	TLT	30-07-2002
Credit*	iShares Corp Bond	LQD	TBD
Common period			2006-02-06

Table: Selected assets for a multi-asset portfolio

*To be implemented after further research on the credit cycle and its impact on the economy.

S&P500 - General description

Characteristics:

- **First** and **largest** US-listed ETF
- Tracks **504 large-cap** US equities
  - float-adjusted market-cap weighting.
- Most **liquid & low-cost** equity instrument
  - with \$600B+ AUM
  - 0.095% expense ratio
- **Highly concentrated** composition
  - Top 10 stocks $\approx$ 37%
  - Tech sector $\approx$ 33%
  - US large-cap only
- SPY is restricted to retail investors under MiFID II/PRIIPs regulation, the UCITS equivalent (SPY5) offers identical exposure within EU compliance.

S&P500 - General description

Top 10 Holdings

#	Name	Weight
1	NVIDIA	7.71%
2	Apple	6.50%
3	Microsoft	5.17%
4	Amazon	3.55%
5	Alphabet (A)	3.09%
6	Broadcom	2.68%
7	Alphabet (C)	2.47%
8	Meta	2.35%
9	Tesla	1.93%
10	Berkshire B	1.58%
Top 10 total		**≈37%**

Sector Breakdown

Sector	Weight
Information Technology	33.34%
Financials	12.29%
Communication Services	10.57%
Consumer Discretionary	9.90%
Health Care	9.51%
Industrials	8.87%
Consumer Staples	5.30%
Energy	3.70%
Utilities	2.53%
Materials	2.02%
Real Estate	1.98%

S&P500 - Historic performances

The **S&P 500 (SPY ETF)** serves as the core equity exposure in the portfolio, designed to boost returns during favorable growth regimes.

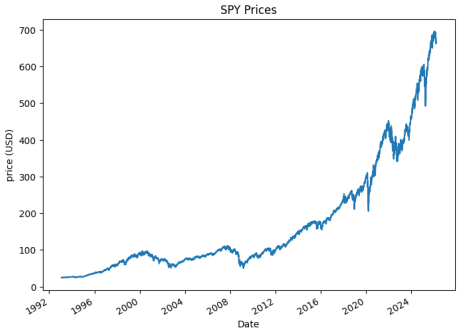


Figure: SPY historical price evolution (1993–2026)

Performance:

- **Period:** 1993–2026 (33 years)
- **CAGR:** 10.54%
- **Annualized Volatility:** 18.6%
- **Sharpe Ratio:** 0.549
- **Sortino Ratio:** 0.779

Insight: Long equity history (33 years) captures multiple regimes and enables robust tail risk estimation.
$\rightarrow$ Sortino $>$ Sharpe indicates that upside volatility is the dominant source of variance.

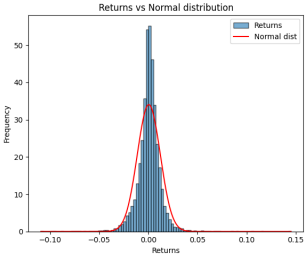
S&P500 - Distribution of returns

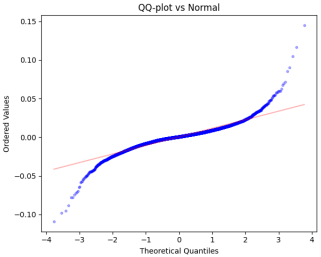
Distribution parameters

Metric	Value
Mean	0.047%
Skewness	-0.010
Kurtosis	11.86
Exc. Kurtosis	8.86

Risk metrics

CL	VaR	CVaR
95%	1.82	2.84
97.5%	2.43	3.59
99%	3.27	4.80





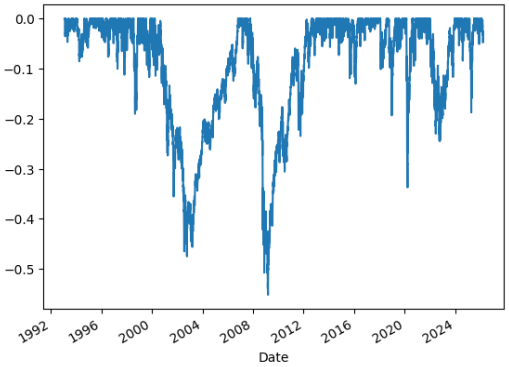
- **Fat tails** confirmed by excess kurtosis: extreme returns occur far more frequently than a normal distribution predicts.
- **Q-Q deviation** at the tails shows that a Gaussian model underestimates crash severity.
- **Near-zero skew** indicates symmetric return distribution: losses and gains are equally likely around the mean.

Insights:

→ Near-zero skew combined with high excess kurtosis indicates that extreme returns are frequent and symmetric: large positive and negative surprises occur with similar probability, but far more often than a normal distribution predicts.

S&P500 - Drawdowns analysis

Historical drawdowns:



Insights: The drawdown profile of SPY spans a wide range of amplitudes, speeds of decline, and recovery times. Four episodes illustrate how different macro drivers produce structurally different drawdown shapes.

- **2000 – Dot-com Bubble:** Longest drawdown in recent history.
- **2008 – Global Financial Crisis:** the deepest episode caused by systemic deleveraging and a prolonged credit shock.
- **2020 – COVID-19 Panic:** a flash crash, followed by a V-shaped recovery within 6 months.
- **2022 – Fed Tightening Cycle:** a moderate but persistent drawdown.

S&P500 - Drawdowns analysis

Longest Dd – Dot-com Bust (2000–2006)

Metric	Value
Peak	Mar 24, 2000
Trough	Oct 9, 2002
Recovery	Oct 26, 2006
Max Drawdown	**-47.52%**
Time Under Water	1,657 days
Recovery Time	1,020 days

Flash Crash — COVID-19 Panic (2020)

Metric	Value
Peak	Feb 19, 2020
Trough	Mar 23, 2020
Recovery	Aug 10, 2020
Max Drawdown	**-33.72%**
Time Under Water	120 days
Recovery Time	97 days

Deepest Dd — Global Financial Crisis (2007–2012)

Metric	Value
Peak	Oct 9, 2007
Trough	Mar 9, 2009
Recovery	Aug 16, 2012
Max Drawdown	**-55.19%**
Time Under Water	1,224 days
Recovery Time	869 days

2022 Rate Hike Cycle — Fed Tightening Shock

Metric	Value
Peak	Jan 3, 2022
Trough	Oct 12, 2022
Recovery	Dec 13, 2023
Max Drawdown	**-24.50%**
Time Under Water	489 days
Recovery Time	294 days

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Macro-systemic Approach to multi-asset investing

Nicolas Poncin is a M.Sc. Financial Engineering student with a strong interest in global macroeconomics and multi-asset portfolio management. He aims to apply his academic background to real investment decision-making. This project was developed as a practical research initiative to explore how macroeconomic regimes influence asset behaviour and portfolio allocation, intending to build a disciplined and testable investment process.



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